Prasiddha Siwakoti

Ph.D. Student
English (Fluent) · Nepali (Native) · Hindi (Fluent)

Morgantown, WV
+1 304-435-2405
prasiddha.siwakoti43@gmail.com
linkedin.com/in/sprasiddha
github.com/agyat4
Google Scholar

EDUCATION

Ph.D. in Electrical Engineering (GPA: 4.0/4.0) *Expected December 2027*
West Virginia University, Morgantown, WV

M.S. in Information and Communication Engineering (83.13%) *April 2020*
Tribhuvan University, Institute of Engineering, Nepal

B.E. in Electronics and Communication Engineering (74.51%) *September 2016*
Tribhuvan University, Himalaya College of Engineering, Nepal

PUBLICATIONS

[1] Eshbaugh, W., Chanana, A., Perez, E., Song, J., Copeland, C.R., Ghadani, S.A., McBride, D., **Siwakoti, P.**, et al. (2026). "Automated Vector-Scanning Spectroscopy for Large-Scale Characterization of Single Quantum Emitters." *Optica* (accepted). [Paper]

[2] **Siwakoti, P.**, Windsor, M., Lewis, C., Segovia-Dominguez, I. (2026). "Quantum Machine Learning for Conflict Fatality and Romantic Relationship Prediction: Entangled with Promise, Measured in Practice." *Lecture Notes in Computer Science*, vol. 16997, pp. 121–130. Springer. [Paper]

[3] **Siwakoti, P.**, Khoshkhahtinat, A., Mehta, P.M., Thompson, B.J., Kirk, M.S.F., da Silva, D. (2025). "Spectral and Spatial Graph Learning for Multispectral Solar Image Compression." *24th IEEE International Conference on Machine Learning and Applications (ICMLA 2025)*. IEEE Xplore. [Paper] [Code]

[4] **Siwakoti, P.**, D'Orazio, V., Segovia-Dominguez, I. (2025). "A Framework for Comparing Vertex-Distribution Graphs: Insights for Social and Conflict Studies." *Lecture Notes in Computer Science*, vol. 16127. Springer. [Paper]

[5] Thrasher, J., Devkota, A., **Siwakoti, P.**, et al. (2025). "Multimodal Federated Learning in Healthcare: A Review." *Journal of Healthcare Informatics Research*. Springer. [Paper]

[6] **Siwakoti, P.** & Ghimire, S.K. (2021). "Devnagari Handwritten Characters Image Super-Resolution based on Enhanced SRGAN." *Journal of the Institute of Engineering*, 16(1), 103–109. [Paper]

EXPERIENCE ~ RESEARCH & ACADEMIA

Graduate Research Assistant | West Virginia University *Jan 2024 – Present*

Multispectral Solar Image Compression *2024 – 2025*
Advisor: Dr. Piyush Mehta (WVU) | NASA Goddard Space Flight Center

- Developed a graph-based learned compression model for NASA SDO solar imagery using spectral graph embeddings (iSWGE) and spatial graph attention (WSGA-C); evaluated on SDOML across six EUV channels. [IEEE ICMLA 2025]

Conflict Fatality Forecasting *2024 – 2025*
Advisors: Dr. Ignacio Segovia-Dominguez, Dr. Vito D'Orazio (WVU) | DRDC Fast Forward Challenge

- Built H2O AutoML models with Tweedie deviance loss for point prediction of armed conflict fatalities and PRIO-grid levels across 1-12 month horizons; derived probabilistic forecasts by parameterizing Tweedie distributions from predicted means.

Evaluation and Diagnostic Framework for Spatiotemporal Forecasting *2025 – Present*
Advisors: Dr. Ignacio Segovia-Dominguez, Dr. Vito D'Orazio (WVU)

- Developing a comparison framework for grids of probability distributions, preliminary work published at [SBP-BRiMS 2025, Springer LNCS]. Journal paper in preparation.

Quantum Dot Emission Analysis *2026*
Advisors: Dr. Ignacio Segovia-Dominguez (WVU), Prof. Edward Flagg (WVU Physics)

- Designed spectral features from quantum dot emission measurements and applied unsupervised clustering to identify emission types. [Optica 2026, accepted]

Instructional Assistant *2026*
West Virginia University | DSCI 101, DSCI 450

AWARDS & ACHIEVEMENTS

- **Winner**, First National FPGA Design Competition, Nepal
- **Merit Scholarship**, Himalaya College of Engineering (B.E.)
- **Merit Scholarship**, Higher Secondary Education Board, Nepal

EXPERIENCE ~ INDUSTRY & DEVELOPMENT

Telecoms Engineer *Oct 2018 – Dec 2023*
Nepal Telecom, Nepal

- *Network Operations & Automation (2021–2023):* Configured and optimized the nationwide backbone network; automated device configuration and recurring tasks using Python (Netmiko, Napalm, Nornir) across hundreds of nodes.
- *Field Engineering (2018–2021):* Led teams for installation and configuration of national radio access infrastructure (BTS, NodeB/eNodeB), microwave and optical fiber transmission systems, and associated power systems.

IP Engineer *Nov 2016 – Apr 2017*
Subisu Cablenet Ltd., Nepal

- Maintained DOCSIS access network operations and configurations.

TEACHING

Part-time Faculty *Oct 2017 – Apr 2018*
Nepal Commerce Campus, Tribhuvan University, Nepal

- Computer Graphics: raster/vector graphics, 2D/3D transformations, shading, and rendering.

Assistant Instructor *Apr 2016 – Oct 2016*
LogicTronix Nepal, Lalitpur, Nepal

- Digital design course for undergraduates: FPGA tools (Xilinx Vivado, ISE), VHDL, Verilog.

PRESENTATIONS

[T1] **Siwakoti, P.** (2026). "Quantum Machine Learning for Conflict Fatality and Romantic Relationship Prediction: Entangled with Promise, Measured in Practice." Presentation, *SBP-BRiMS 2026*.

[T2] **Siwakoti, P.** (2025). "Spectral and Spatial Graph Learning for Multispectral Solar Image Compression." Oral presentation, *24th IEEE International Conference on Machine Learning and Applications (ICMLA 2025)*.

[T3] **Siwakoti, P.** (2025). "A Framework for Comparing Vertex-Distribution Graphs: Insights for Social and Conflict Studies." Presentation, *SBP-BRiMS 2025*.

[T4] **Siwakoti, P.** (2025). "Spectral Graph Embeddings for Multispectral Solar Image Compression." Poster presentation, *WVU AI Symposium: AI in Action*, West Virginia University, May 2025.

SERVICE

- **Peer Review:** Reviewed 4 manuscripts for SBP-BRiMS 2025 and 2026.
- **Judging:** Evaluated undergraduate research posters across engineering and computing disciplines.

TECHNICAL SKILLS

Languages	Python, C/C++, R, JavaScript
ML/DL	PyTorch, TensorFlow, Keras, scikit-learn, H2O AutoML
Quantum Tools	Qiskit, Quantum Kernel Methods, Variational Quantum Classifiers & Neural Networks, Quantum Data Encoding, Noiseless Statevector Simulation
Libraries	NumPy, Pandas, Matplotlib, Seaborn, OpenCV
Architectures	Graph Neural Networks (GNNs), CNNs, GANs, VAEs, Transformers
Math & Stats	Topological Data Analysis, Multivariate Statistics, Linear Algebra, Statistical Inference, Calculus and Optimization
Tools	Git/GitHub, Hugging Face, Streamlit